KRISHNA CLASSES

WORKSHEET

SUBJECT- MATHS

CHAPTER- 4

NAME OF STUDENT –

ISSUE DATE –

MCQ

1. The general form of a quadratic equation is:
   a. $ax^2+bx+c=0$
   b. $ax+bx^2+c=0$
   c. $x^2+bx+c=0$
   d. $ax^2+b+c=0$
2. A quadratic equation has how many solutions?
   a. 0
   b. 1
   c. 2
   d. 3
3. The sum of a quadratic equation's will be:
   a. –b/a
   b. c/a
   c. –c/a
   d. b/a
4. The product of the roots of a quadratic equation is:
   a. –b/a
   b. c/a
   c. –c/a
   d. b/a
5. What is a quadratic equation's discriminant?
   a. b2−4ac
   b. −b/a
   c. c/a
   d. –c/a
6. If the discriminant of a quadratic equation is negative, then the roots are:
   a. Real and distinct
   b. Real and equal
   c. Imaginary Numnber
   d. Not defined
7. The roots of the quadratic equation $2x^2+5x+3=0$ are:
   a. −3/2 and −1
   b. −1/2 and −2/3
   c. −1/2 and −3/2
   d. −1/3and −3/2
8. If the roots of a quadratic equation are real and equal, then the discriminant is:

a. positive

b. Negative

c. Zero

d. Not defined

9. The quadratic equation $x^2+2x+1=0$has:
   a. No real roots
   b. Two real and equal roots
   c. Two real and distinct roots
   d. One real and one complex root
10. The quadratic equation $x^2-6x+9=0$ has:
   a. No real roots
   b. Two real and equal roots
   c. Two real and distinct roots
   d. One real and one complex root
11. The quadratic equation $x^2-4x+4=0$ has:
   a. No real roots

   b. Two real and equal roots

   c. Two real and distinct roots

   d. One real and one complex root
12. The quadratic equation $3x^2-5x+2=0$has:
   a. No real roots

   b. Two real and equal roots

   c. Two real and distinct roots

   d. One real and one complex root
13. The roots of the quadratic equation $4x^2-8x+3=0$are:
   a. $(8\pm4)/8$
   b. $3/2,1/2$
   c. $1/2, \frac{\sqrt{5}}{2}$
   d. $2/3, \frac{\sqrt{7}}{3}$

14. The value of p for which the quadratic equation $x^2+px+9=0$ has equal roots is:
   a. ±3
   b. ±2
   c. ±6

d. ± 8

15. The nonzero roots of the equation $x^2+3x+k=0$ are in the ratio of 2:1. What is the value of k?
    a. 0
    b. 3
    c. 2
    d. 1

16. Find the roots of the equation $x^2 - 3x - m(m + 3) = 0$, where m is a constant
17. If $x = -\frac{1}{2}$, is a solution of the quadratic equation $3x^2 + 2kx - 3 = 0$, find the value of k.
18. Find the value of p so that the quadratic equation $px(x - 3) + 9 = 0$ has two equal roots.
19. Find the roots of $4x^2 + 3x + 5 = 0$ by the method of factorization.
20. Solve for x:
    $36x^2 - 12ax + (a^2 - b^2) = 0$
21. Find the value of k for which the equation $x^2 + k(2x + k - 1) + 2 = 0$ has real and equal roots
22. Solve the following quadratic equation for x: $4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0$
23. Solve the following for x: $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$
24. If $x = \frac{2}{3}$ and $x = -3$ are roots of the quadratic equation $ax^2 + 7x + b = 0$, find the values of a and b.
25. Solve for x: $\sqrt{6x + 7} - (2x - 7) = 0$

26. Solve the equation $\frac{3}{x+1} - \frac{1}{2} = \frac{2}{3x-1}$

27. Three consecutive natural numbers are such that the square of the middle number exceeds the difference of the squares of the other two by 60. Find the numbers.

28. If the sum of two natural numbers is 8 and their product is 15, find the numbers.

29. Find the roots of the equation $\frac{1}{x+4} - \frac{1}{x-7} = \frac{11}{30}$.

30. A shopkeeper buys some books for 80. If he had bought 4 more books for the same amount, each book would have cost ₹1 less. Find the number of books he bought.

31. The diagonal of a rectangular field is 16 meters more than the shorter side. If the longer side is 14 meters more than the shorter side, then find the lengths of the sides of the field.

32. The sum of two numbers is 9 and the sum of their reciprocals is $\frac{1}{2}$. Find the numbers.

33. The numerator of a fraction is 3 less than its denominator. If 1 is added to the denominator, the fraction is decreased by $\frac{1}{15}$. Find the fraction.

34. A train travels 180 km at a uniform speed. If the speed had been 9 km/hour more, it would have taken 1 hour less for the same journey. Find the speed of the train.

Case study

Raj and Ajay are very close friends. Both the families decide to go to Ranikhet by their own cars. Raj's car travels at a speed of x km/h while Ajay's car travels 5 km/h faster than Raj's car. Raj took 4 hours more than Ajay to complete the journey of 400 km.

1. What will be the distance covered by Ajay's car in two hours?
   a. $2(x + 5)$km
   b. $(x - 5)$km
   c. $2(x + 10)$km
   d. $(2x + 5)$km
2. Which of the following quadratic equation describe the speed of Raj's car?
   a) $x^2 - 5x - 500 = 0$
   b) $x^2 + 4x - 400 = 0$
   c) $x^2 + 5x - 500 = 0$
   d) $x^2 - 4x + 400 = 0$
3. What is the speed of Raj's car?
   a) 20 km/hour
   b) 15 km/hour
   c) 25 km/hour
   d) 10 km/hour

4. How much time took Ajay to travel 400 km?

a) 20 hour
b) 40 hour
c) 25 hour
d) 16 hour